

PhD Progress Review

Reusable human judgment for agentic-AI-driven variability exploration

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Wednesday, 12 August 2026, 09:00 · presenter: Ali Hamed

Working draft — research-question wording is provisional pending your confirmation today; nothing here implies prior approval.

What we will decide today

- **Confirm the research-question architecture — one umbrella RQ and exactly three sub-questions (the decision this meeting exists for).**
- Walk through the Gap & Research-Questions chapter, showing every correction from our 5 August working call is incorporated.
- Show the per-question literature coverage and where the gaps are.
- State the current evidence honestly — what is established, and what is not yet.
- Confirm the plan structure and the immediate next task.

Record note: this progress package is built from a machine transcript of the 5 August call (speakers inferred, no diarization). Wordings are reconstructed and provisional until you confirm them here.

Progress since the 5 August call

Item	Status
Gap & Research-Questions chapter — full draft written	In progress
RQ + SQ1–SQ3 wording migrated across proposal & governance docs	In progress
Per-question literature workbook + coverage-gap analysis	In progress
Artifact-per-SQ options considered (thinking only, per your instruction)	Done
Weekly Wednesday 09:00 cadence confirmed by all three of us	Done
Verify final wording against my saved call draft	Open / blocked
Share the project Drive with you both	Open / blocked

"In progress" = drafted, pending your sign-off. "Open" = my action this week. No item is marked complete on the basis of a draft alone.

One umbrella question, exactly three sub-questions

Umbrella RQ. How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human–AI co-reasoning?

SQ1 — Selective intervention

When and how, in variability exploration scenarios, should an agentic assessment system request human judgment so that important uncertainties are addressed without unnecessary expert burden?

SQ2 — Governed knowledge reuse

How should expert judgment — including the system's core reasoning — be represented, validated, reconciled, and stored so it can be reused transparently without unsafe generalization or loss of human authority?

SQ3 — Evaluation and transfer

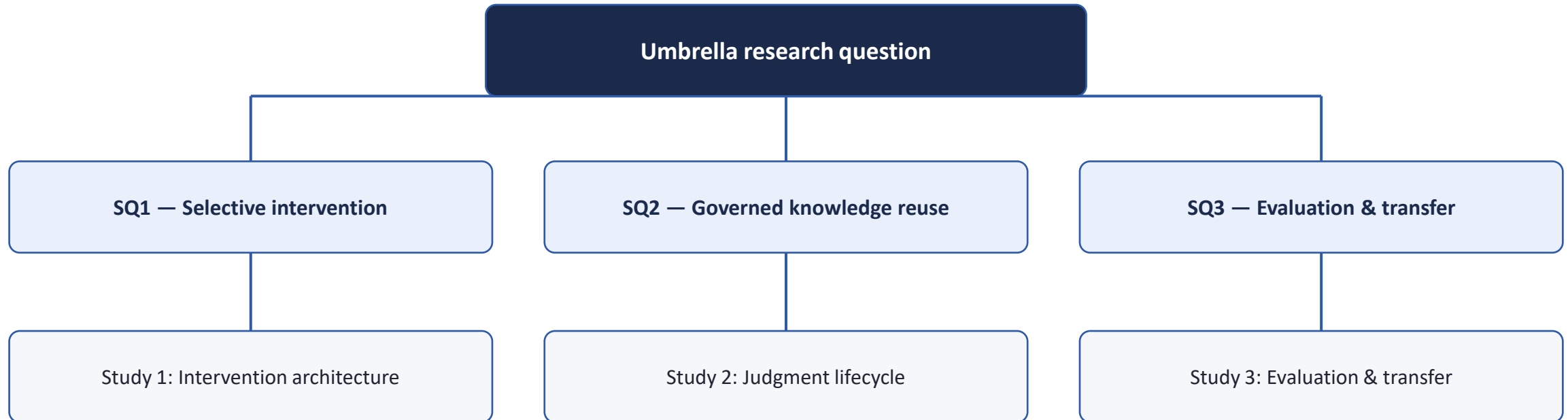
How can expert judgment be reused and transferred across different guideline-operationalization contexts without unsafe generalization or loss of human authority, first in software/modeling and, when governance and access permit, in healthcare?

Verbatim wording shown for sign-off. Decision requested: D-RQ-01 (umbrella) and D-RQ-02 (SQ1–SQ3) — Confirm / Confirm with correction / Retire / Defer.

Every correction from 5 August is incorporated

From the call	How it is reflected
Arnon: separate the solution from the research question; cut padding	Gap section argues why the question is open — it does not pitch the agentic design
Iris: drop "reused" from the headline	Reuse moved into SQ2/SQ3; headline is about supporting the activity
Iris: narrow to variability exploration	"variability exploration in guideline operationalization" named explicitly
Iris/Arnon: keep "reliable", lean off "auditable/transferable/end-to-end"	Headline trimmed to "reliable" (leaning — for your confirmation)
Arnon/Iris: split SQ2 into capture vs. future-transfer; prefer "expert"	SQ2 spans both halves; "expert judgment" used (open choice, leaning expert)
Iris: tie SQ2 to "core reasoning" + evaluation clause	SQ2 names core reasoning and carries a balance/authority clause
Arnon: SQ3 classifies domain-specific vs. transferable uncertainty	SQ3 rationale built on that classification (actor/use-case = general gap)
Iris: keep RQs domain-neutral; domains live in methodology	RQ text is domain-neutral; SE/medical framing deferred to Chapter 4

Each sub-question maps to one study



Full contract (method, evidence, artifact, metrics, risk, fallback, publication target per study): [three-study-contract.md](#).

Decision requested: D-RQ-03 — study mapping and method direction.

Per-question literature: honest coverage

- Tagging is in place — every source is labelled RQ1 / RQ2 / RQ3 / general (native Excel workbook delivered).
- RQ1 (selective intervention): thin — one survey; agentic-specific oversight work still needed.
- RQ2 (governed reuse): tool-heavy, research-light — capture/provenance/RLHF-contrast papers missing.
- **RQ3 (evaluation & transfer): currently empty — the highest-priority gap in the map.**
- Each gap is mapped to the frozen query (QL-01..05) or explicit snowballing that will close it.
- No database search has been run yet — so no novelty or coverage-completeness claim is made.

Decision requested (time permitting): D-RQ-08 — initial literature scope and taxonomy direction.

Current evidence — what is, and is not, established

Established

- Mechanism & architecture: capture, memory, advisory, and non-destructive comparison are built and tested (M1–M4B-1).
- Traceability, explainable retrieval, review routing, governance controls.
- Readiness to run the leakage-controlled evaluation.

Not yet established

- Accuracy, macro-F1, generalization, effort reduction, or superiority.
- **EXP-005: 0 labels supplied (27 blind rows, 24 generalization-safe; gate needs ≥ 20).**
- Any clinical or medical result (medical readiness 0 / 6 gates; MIMIC not selected).

Decision requested: D-RQ-07 — evidence-boundary wording and the five claim states.

Plan structure and horizon

- **Common core (both plans): complete Studies 1 & 2 and evaluate the framework first in software/modeling.**
- Plan A — conditional medical extension: only after all six gates (use-case, people, authorization, ethics/privacy, environment, protocol) pass.
- Plan B — complete software/modeling route: a second authorized SE/modeling context; answers every RQ without medical data.
- **Fallback control: 26 August internal checkpoint (not supervisor-approved) — if any gate lacks an owner/evidence/date, Plan B becomes the committed September route.**
- Plan is written in ~3-month, semester-aligned blocks over a 3-year horizon — not month-by-month.

Decisions requested: D-RQ-04 (Plan A/B), D-RQ-05 (every RQ answerable under Plan B), D-RQ-06 (26 Aug date).

Decisions requested today

#	Decision	Priority
D-RQ-01	Umbrella research-question wording	Mandatory
D-RQ-02	SQ1, SQ2, SQ3 wording	Mandatory
D-RQ-03	Three-study mapping & method direction	Mandatory
D-RQ-04	Plan A conditional / Plan B complete	Mandatory
D-RQ-05	Every RQ answerable under Plan B	Mandatory
D-RQ-07	Evidence-boundary wording & 5 claim states	Mandatory
D-RQ-06	26 August fallback control date	Time permitting
D-RQ-08	Initial literature scope & taxonomy	Time permitting
D-RQ-09	Bounded metadata-only MIMIC boundary	Time permitting
D-RQ-10	Owners: medical feasibility & university process	Assignment

If time is short, close the six Mandatory rows and assign owners for the rest.

Immediate next steps

- **This week: finalize the confirmed wording, finish the Chapter-3 draft, complete the per-RQ literature tagging in the shared Sheet, share the Drive.**
- Think about (not yet start) the literature survey and the research artifact per SQ.
- Maintain the Word proposal plus a separate tracking document.
- Next meeting: the following Wednesday under the confirmed weekly 09:00 cadence — goal: Chapter 3 complete and reviewed.

Close-out: read back today's decisions, then agree exactly one primary next task with an owner, a date, and a definition of done.

What this package does not claim

- The research-question wording is reconstructed and provisional until confirmed today.
- No literature search has been executed; no novelty or coverage-completeness claim.
- EXP-005: 0 labels supplied (27 blind rows, 24 generalization-safe; gate needs ≥ 20) — no accuracy/generalization/effort claim.
- Medical readiness 0 / 6 gates; MIMIC not selected; no elapsed-hours audit claim.
- The Drive is not shared/access-tested until it actually is.
- No requirement is complete/accepted; 26 Aug is internal; Sept/Oct are not official university deadlines; no partner access without written evidence.
- The transcript is machine-derived: no verbatim quotes, no confident attribution of unattributed segments.

Thank you

Ready to confirm the research-question architecture and set this week's one next task.